Product Design

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Problem Statement

Design a search product for a healthcare platform.

Clarifying Questions

1. **Why we are designing the platform or What is the platform's primary purpose?**

   The product will provide a easy user roadmap as existing user journey of a healthcare seeking persona is very difficult . one search solution which will be efficient and easy.
2. **Are we targeting users across both app and web platforms, or is the focus on one specific platform?**

   We are targeting users across both app and web platforms, with a mobile-first approach."

3. **What kind of users are we focusing on?**

   Patients seeking healthcare service and other medical information.

4. **Is the target audience limited to users within India, or are we considering a global user base?**

   Initially, we are focusing on users in India.

5. **What's the key problem we're solving with the search feature?**

   Users often face difficulty to find the right doctors, services, or articles quickly and easily.

6. **Do we have any resource constraints (e.g., development time, budget, team size) that might impact our approach?**
   Yes, we have limited development resources, limited engineering bandwidth and a strict deadline of 12 months.

7. **Are there any accessibility requirements, such as multi-language support to address inclusivity?**

   Accessibility is crucial, especially multi-language support for regional users.

8. **What role does accessibility play in the design?**

Accessibility is critical ,like voice search especially for elderly users and those with lower digital literacy.

9. **What is the expected outcome or key success metric for this search product?**

The key metric/north star metric is search-to-booking conversion rate

other metrics - Search accuracy, search time, user satisfaction, and re-appointment bookings

10 . **Can we leverage any existing strengths?**

We do have user behavior analytics, including common search terms and drop-off points. We also have a strong database of healthcare providers and medicines.

Competitive Analysis

Market Trends

- **Telemedicine Expansion:**Telemedicine continues to be a major trend, bridging the gap between patients and healthcare providers. Its adoption is expected to grow as it offers convenient access to medical services, particularly in rural areas
- **Digital Adoption**:Increasing smartphone penetration (800M+ users).Growing internet connectivity in Tier 2/3 cities.Rising digital literacy among older population
- **Digital Health Records and Prescriptions:**The use of digital prescriptions (E-Rx) and electronic health records is on the rise, enhancing patient care by reducing errors associated with paper-based systems. This trend is part of a broader move towards a paperless healthcare environmentRise of voice-enabled search in healthcare apps.

- **Government Support and Reforms:**Favorable government policies and reforms aimed at enhancing digital health infrastructure are expected to further propel market growth.India is seeing rapid digital adoption in healthcare, driven by the rise of telemedicine and government initiatives like Ayushman Bharat Digital Mission.
- **Market Growth Projections:**The mHealth apps market in India is projected to grow at a CAGR of 17.1%, reaching approximately USD 4.83 billion by 2030, driven by increased smartphone penetration and rising health awareness among consumers**.**
- **Chronic Disease Management:** Apps specifically designed for managing chronic conditions like diabetes, hypertension, and asthma, offering personalized plans, medication reminders, and data tracking
- **Telemedicine Adoption:** The COVID-19 pandemic has accelerated the adoption of telemedicine services. The telemedicine market in India is expected to grow at a CAGR of 24% from 2023 to 2030.
- **Digital Health Integration:** The integration of digital health technologies, including mobile health apps and wearable devices, is transforming healthcare delivery in India. These tools facilitate health tracking, medication adherence, and lifestyle management, empowering individuals to actively manage their health.
- **AI & Machine Learning**: AI is being integrated for tasks like:

    1. Disease prediction & diagnosis: Algorithms analyze patient data to identify potential health risks.
    2. Drug discovery: AI accelerates drug development and personalized medicine.
    3. Chatbots: AI-powered chatbots provide initial patient triage, answer basic health queries, and schedule appointments.

Key Competitors:

	Strength	Weakness
Practo	Strong in doctor discovery and teleconsultation	medicine delivery and lacks personalization.
PharmEasy	Focused on discounted medicines	limited diagnostics, and no doctor discovery
Apollo 24/7	Reliable with hospital-backed services	higher costs and lacks user-friendly design.

Zocdoc	Focuses on finding doctors by location, specialty, and availability. It's great for booking appointments	doesn't offer a full view of health records or insurance details.
Healthgrades	Helps people find doctors and healthcare providers based on reviews and ratings	it doesn't connect with health records or offer personalized recommendations.

GAP Analysis :

Market Gaps

While these platforms are useful, they have some key areas where they fall short:

- **Limited personalization of search results**: Users are not provided with search results tailored to their health history or preferences.
- **Interoperability Issues:**Lack of interoperability between different healthcare systems can hinder seamless data exchange, affecting patient care continuity.
- **Poor discoverability of niche services (e.g., rare specialties)**: Specialized services and doctors are often hard to find, making it difficult for users to locate the right care.
- **Lack of robust filtering options for insurance and language**: The absence of filters for specific insurance plans or regional language preferences limits accessibility for users with specific needs. Need for **multilingual support** to serve diverse linguistic groups in India.
- **Ineffective symptom-to-doctor mapping**: Users often struggle to find the right doctor for their symptoms, as the search product doesn't effectively map symptoms to specialists.
- **Interoperability Issues:**Lack of interoperability between different healthcare systems can hinder seamless data exchange, affecting patient care continuity.
- **Lack of Preventive Focus**: Few proactive tools like symptom checkers or fitness tracking.

These gaps present an opportunity to develop a healthcare search platform that **offers more personalized results**, **better discoverability for niche services**, *and* **enhanced filtering for insurance** and **language preferences**, *ultimately improving the overall user experience.*

User Segmentation:

SEGMENT	AGE	INCOME	BEHAVIOUR	CHALLENGES	MOTIVATION
HEALTH-CONSCIOUS YOUNG PEOPLE	18-28years	Low-to-mid (students, early-stage professionals).	Look for preventive medical care (diet, exercise). Active on health apps; occasional teleconsultations.	restricted money available for spending. locating reliable and reasonably priced services.	maintaining one's fitness. The ease of remote medical care.
URBAN TECH-SAVVY BUSY PROFESSIONAL	28-45 years	Mid-to-high (stable income).	Look for dependable, reliable online consultations. To ensure regular inspections for symptom checking, schedule lab tests.	insufficient time for face-to-face meetings. Health is frequently neglected.	Convenience and effectiveness. sustaining optimal performance at work.
URBAN FAMILES	35-50 years	Mid-to-high (dual-income households).	Oversee the elderly and children's medical care. Make use of subscriptions for medications. regular consultations for long-term care.	arranging for several dependents' care. balancing quality and price.	ensuring the welfare of the family. budget orientation for family.

ELDERLY PEOPLE AND RETIREES	55+ years	Mid-to-low (retirement savings, pensions).	Pay attention to managing chronic illnesses. standard diagnostic procedures. lab tests and medicine on time.	obtaining long-term care and prescription drugs. Complex user experience	Home-based services that are convenient. preserving one's independence and standard of living.
RURAL USERS	All users	Low-to-mid	Limited healthcare options. Unavailability of specialist doctors. Own regional language availability.	Cost effective treatment. Low digital literacy. Awareness of available services	Affordable, accessible healthcare treatment. Specialist doctors. Multilingual support.

Prioritized User Segment

We should prioritize Busy Professionals and families because:

•**Business Growth**: Urban areas have growing healthcare needs but supply access to specialized care is limited. The priority for the urban professional is highest due to the frequency of the usage of the platform
•**Revenue and Frequency**: Give top priority to market segments that provide reliable and worthwhile interactions.
• **Scalability**: Prioritize tech-savvy urban users over intricate, low-revenue rural markets.
• **Alignment with Business Objectives:** Family-focused services provide recurrent income, while reaching out to young adults.

User Persona: Urban busy Professional with family

Persona : Tarun Pal

Age	35
Occupation	Technical manager
Tech Proficiency	Moderate
Income	₹15-20 LPA
Location	Bangalore
Family status	Married (Old parents and one child)

Pain points

Pain points	Goals	Motivations
Balancing health with work-life commitments. **appointment availability struggles with language barriers.** **Difficulty in finding nearby child doctors or quick emergency care.** **Getting medicine and medical diagnostics easily**	Save time as per busy schedule. Reliable healthcare Schedule an appointment without hassle. Ensure safety related to family and personal health seamlessly	No language issues Quick healthcare service Time-saving solutions Preventive healthcare and lab tests Proactive reminders.

User Journey for Tarun Pal

Awareness

Current Experience:

1. Appointment scheduling – Appointment scheduling for a consultation is a time taking task most of the time need to visit offline clinics and hospitals but the appointment is guaranteed.

2. Quality of treatment – As most of the first timers do not know about the quality of the treatment of the doctor

3. Word-of-mouth recommendations for doctors – Mostly user rely on the word-to-mouth recommendations for doctors.

4. History data filling –History medical data feeding and other health reports need to be carried manually.

5. Other healthcare services scheduling –Scheduling medicines and lab tests are difficult activities

6. Searching Healthcare services- Searching healthcare services depends on basic google search or asking friends and relatives.

Proposed Solution: Marketing campaigns and advertisements on social platforms showcasing easy doctor search and other healthcare services.

Research

Current Experience:

- Searches on Google with prompt e.g. Fine Doctor near me.
- Going to the offline clinic or hospital checking details
- Asking friends or relatives.

Proposed Solution:

- **Search Bar with Auto-Complete:** Suggestions based on user input, such as "Cardiologist in Bangalore."
- **Auto-suggestion – The search should be enabled with auto suggestion and auto complete feature.**
- **Multilingual Filters:** Drop-down menu for languages (e.g., Hindi, Tamil).
- **Voice Search:** Allow input in local languages for users with low literacy.

Appointment Booking

Availability – Calling the clinics manually , struggles to get a time slot for appointment.

Time constraint- Appointment is scheduled as per time availability of the clinic or the hospital and getting options very difficult .May need to take leave or time off work to visit.

Quality of the Healthcare - Difficult to decide the quality of the doctor ,unclear on which doctor to trust.

Rescheduling of the appointment - Mostly there is no option for rescheduling of appointment in the clinics ,we need to cancel the bookings and make a fresh booking.

Proposed Solution:

Real time slot booking -Real-time availability and instant booking confirmations.

Rating and reviews - Display verified ratings and reviews with no of patients trusted count.

Rescheduling with other options- Option to reschedule easily without penalties and provide other options in same category .

Consultation

Unable to visit the offline clinics due age/disability/time constraint.

Online consultation -Option to choose communication mode (video, audio, or chat).

Lab tests and diagnostics-

Provide option schedule for lab tests at home as medical technician will visit and the samples will be taken from home .

Scheduling of the Lab based tests- scheduling option for Lab based tests like MRI , X-ray .

Post Consultation

Medical history card- Manages the prescriptions document as , no options for a digital medical history .

Medicines- Visits a pharmacy or medicine shop to purchase medicines , often facing delays or unavailability

Post consultation follow-up – No scheduling for follow-up check-ups.

Rate the service and voice of customer- No scope for the customer to rate and review the treatment he received.

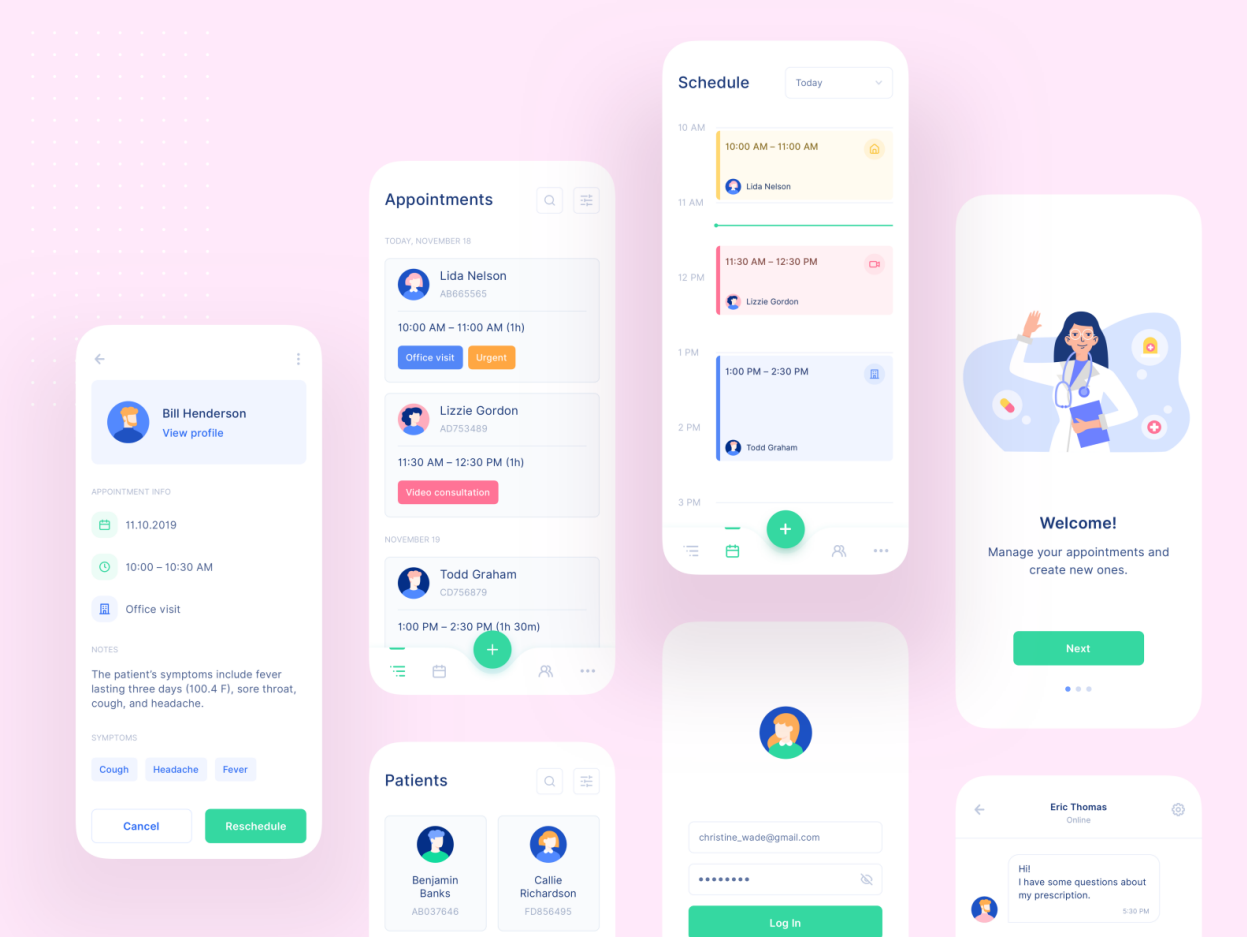
Proposed Solution:

- Reminder notifications- notifications for follow-up check-ups.
- Feedback/reviews section- Customer feedback and reviews after the successful consultation.
- Medical history data and prescriptions database- Medical history and the prescriptions need to be stored in the patients profile.
- Medicine store and delivery- medicine store should be integrated and delivery of the medicines as pre schedule.

Pinpoints Prioritization

Feature Description	MoSCoW Category	Justification
1. Unreliable, inefficient searches for quality healthcare	Must Have	Essential for user satisfaction; without reliable searches, users cannot find necessary services.
2. Slot availability and appointment booking – struggles to get a time slot	Must Have	Critical for user experience; users need to secure appointments easily to access healthcare.
3. Time constraints in finding time for in-person consultations	Should Have	Important but not critical; while it enhances usability, users can still consult online if needed.
4. Quality of healthcare – reviews, certificates, and badges	Should Have	Important for building trust; enhances user confidence in choosing providers but not immediately critical.
5. Unable to visit offline clinics – online consultation	Must Have	Essential for accessibility; provides necessary healthcare options when physical visits aren't possible.
6. Inefficient healthcare experience (long wait times, multiple visits)	Should Have	Important for improving overall user experience; reduces frustration but not essential at launch.
7. Healthcare management challenges (forgetting follow-ups, misplaced prescriptions)	Must Have	Critical for continuity of care; proactive reminders are essential for effective healthcare management.

8. Cost transparency challenges (unknown fees for treatments)	Should Have	Important for user trust and budgeting; enhances user experience but can be addressed later if needed.



Research and Development Phase

1. **User Interviews**:
    - Conduct detailed user interviews in semi-urban areas to understand the needs and preferences of potential users. This helps in creating user personas and identifying key features for the app.
2. **Prototyping**:
    - Build prototypes focusing on multilingual search capabilities and real-time availability of healthcare services. This step is crucial for validating ideas before full-scale development.
3. **Minimum Viable Product (MVP) Launch**:

- Aim to launch the MVP by Month 6, incorporating essential features that address core user needs while allowing for future iterations based on feedback.

4. **Iteration and Scaling**:
    - Post-MVP launch, gather user feedback to inform subsequent iterations. Focus on scaling the app's functionalities based on user engagement and market demands.

Product Design Features

1. **Search Interface**:
    - Include auto-completion features to enhance search efficiency.
    - Support voice search with multilingual capabilities.
2. **Categories and Filters**:
    - Primary categories: Appointments, Medicines, Lab Tests.
    - Secondary categories: Different specialist doctors.
    - Filters based on specialties, distance, and fees.
3. **Language Selection**:
    - Enable language selection to cater to regional populations.
4. **Scheduling and Booking**:
    - Streamlined booking process integrated within the app interface.
5. **Doctor Profile Page**:
    - Comprehensive profiles featuring reviews and ratings from previous patients.
6. **Feedback Page**:
    - A dedicated section for users to provide feedback on their experiences.
7. **Telemedicine and E-Pharmacy Orders**:
    - Facilitate telemedicine consultations and e-pharmacy orders directly through the app.
8. **Post Consultation Check-Ups**:
    - Ensure follow-up care through reminders for check-ups or additional consultations as needed.

Feature Prioritization (RICE Scoring)

Feature	Reach	Impact	Confidence	Effort	RICE Score Calculation	RICE Score
Search Interface	5000	2	0.8	3	(5000×2×0.8)/3(5000×2×0.8)/3	2666.67
Categories and Filters	4000	1	0.9	2	(4000×1×0.9)/2(4000×1×0.9)/2	1800
Language Selection	3000	1	0.7	1	(3000×1×0.7)/1(3000×1×0.7)/1	2100
Scheduling and Booking	6000	3	0.9	4	(6000×3×0.9)/4(6000×3×0.9)/4	4050
Doctor Profile Page	4500	2	0.85	3	(4500×2×0.85)/3(4500×2×0.85)/3	2550
Feedback Page	2000	1	0.6	2	(2000×1×0.6)/2(2000×1×0.6)/2	600
Telemedicine and E-Pharmacy Orders	7000	3	0.95	5	(7000×3×0.95)/5(7000×3×0.95)/5	3975
Post Consultation Check-Ups	3500	2	0.75	3	(3500×2×0.75)/3(3500×2×0.75)/3	1750

Prioritization

Based on the calculated RICE scores, we can prioritize the features as follows:

1. **Scheduling and Booking: RICE Score = 4050**
2. **Telemedicine and E-Pharmacy Orders: RICE Score = 3975**
3. **Search Interface: RICE Score = 2666.67**
4. **Doctor Profile Page: RICE Score = 2550**
5. **Language Selection: RICE Score = 2100**
6. **Categories and Filters: RICE Score = 1800**
7. **Post Consultation Check-Ups: RICE Score = 1750**
8. **Feedback Page: RICE Score = 600**

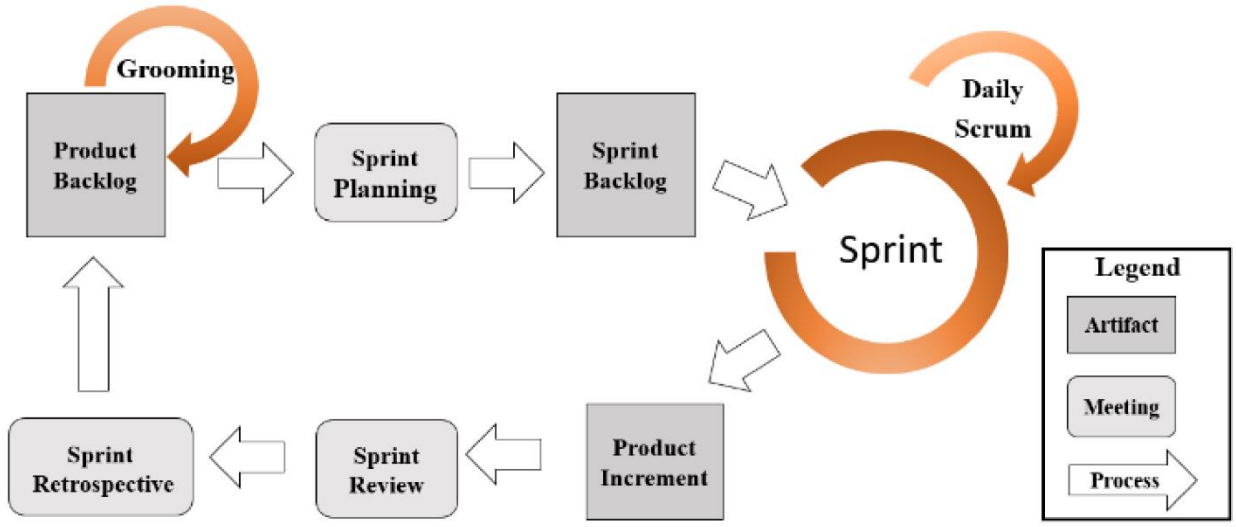
User Design Roadmap :

1. **User Opens the App:**
    - **Design:** Clean and minimalistic interface with a prominent search bar and voice search button.
    - **Development:** Implement search bar with auto-completion and voice search integration.
2. **User Prompt:**
    - **Design:** Clear and concise search prompts (e.g., "Find a doctor," "Book an appointment").
    - **Development:** Implement natural language processing for better search query understanding.
3. **Search Results:**
    - **Design:** Display search results in a user-friendly format with filters for specialty, distance, availability, and reviews.
    - **Development:** Integrate location services and implement efficient filtering mechanisms.
4. **Doctor Profile View:**
    - **Design:** Detailed profile with information on qualifications, experience, fees, availability, reviews, and consultation options.
    - **Development:** Integrate with doctor databases and enable user reviews and ratings.
5. **Appointment Confirmation:**
    - **Design:** Clear appointment confirmation screen with details, payment options, and reminders.
    - **Development:** Implement secure payment gateway integration and reminder notifications.
6. **Post-Appointment:**
    - **Design:** Reminders for follow-up appointments and review submissions.
    - **Development:** Implement automated reminder system with customizable timings.
7. **Post-Consultation:**

- **Design:** Options for ordering medicines, scheduling lab tests, and providing feedback.
- **Development:** Integrate with e-pharmacy and diagnostic lab services.

8. **Medicine Ordering:**
    - **Design:** Options for manual or automated medicine ordering based on prescriptions.
    - **Development:** Implement secure prescription uploading and order tracking.
9. **Medical History Check:**
    - **Design:** Secure access to previous prescriptions and lab reports.
    - **Development:** Implement secure data storage and retrieval mechanisms.

Product Development Planning-



Design Sprint Phases:

Phase 1: Discovery & Research (Months 1-3)

- **Sprint 1-2:** User Research
    - Conduct in-depth user interviews in semi-urban areas to understand their healthcare needs, pain points, and preferred communication channels.
    - Analyze user data and create user personas and empathy maps.
- **Sprint 3-4:** Competitive Analysis
    - Research existing healthcare apps in the market, identifying their strengths and weaknesses.
    - Analyze competitor features, user reviews, and market trends.

- **Sprint 5-6:** Information Architecture & Wireframing
    - Define the core functionalities and user flows for the app.
    - Create low-fidelity wireframes to visualize the user interface and user experience.

Phase 2: Design & Development (Months 4-9)

- **Sprint 7-10:** Prototyping & Usability Testing
    - Develop interactive prototypes of key features (search, doctor profiles, appointment booking).
    - Conduct usability testing with target users to gather feedback and iterate on the design.
- **Sprint 11-16:** Development & Integration
    - Begin development of the core app features (search, user profiles, basic appointment scheduling).
    - Integrate with relevant APIs (e.g., location services, payment gateways).
- **Sprint 17-20:** Multilingual Support & Accessibility
    - Implement multilingual search and user interface.
    - Ensure accessibility for users with disabilities (e.g., screen reader compatibility).

Phase 3: Launch & Iteration (Months 10-12)

- **Sprint 21-22:** MVP Launch & Monitoring
    - Launch the Minimum Viable Product (MVP) with core functionalities.
    - Monitor app usage, user feedback, and key performance indicators (KPIs).
- **Sprint 23-24:** Iteration & Scaling
    - Based on user feedback and data analysis, prioritize and implement improvements.
    - Expand features (e.g., telemedicine, e-pharmacy, medical history integration).
    - Scale the app to reach a wider audience through marketing and partnerships.

Key Features & Design Considerations:

- **User-Centric Design:** Prioritize user needs and preferences throughout the development process.
- **Intuitive Search:** Implement robust search functionality with auto-completion, voice search, and multilingual support.

- **Doctor Profile Transparency:** Display comprehensive and verified doctor profiles with reviews and ratings.
- **Seamless Appointment Scheduling:** Enable easy and flexible appointment booking with real-time availability.
- **Personalized Experience:** Tailor the app experience to individual user preferences and needs.
- **Data Security & Privacy:** Ensure the security and privacy of user data through robust security measures.